ArtISMo's Final EVENT AGENDA

Programme du Final Event ArtISMo

ArtISMo Final EVENT Agenda

June 15th and 16th, 2026

ESIGELEC ROUEN, Bâtiment TURING

Amphi Robert VALLEE

ArtISMo: Intelligent Estimation Algorithms for Smart Mobility

Monday, 15 June 2026

10h-11h	Final Event's Coffee Reception and Registration	
11h-12h	ANR ArtISMo's Presentation	
	11h - 11h15	ArtISMo Final Event Welcome: J. Le GUEN (ESIGELEC Director)
	11h15 - 11h30	ArtISMo Final Event Welcome: R. KHEMMAR (ArtISMo Project Manager)
	11h30 - 12h15	ArtISMo Project Synthesis: A. ZEMOUCHE (ArtISMo Lead Partner)
12h15-13h30	Lunch Break	
13h30-15h10	**ESIGELEC Partner's Presentation**	
	13h30 - 13h50	R. KHEMMAR: IRSEEM's Contribution in ArtISMo (Synthesis)
	13h50 – 14h10	F. JENDOUBI: Perception of Complex Scenes for Smart Mobility based on Deep Learning and Multi-Task Learning (MTL)
	14h10 – 14h30	L. LECROSNIER: Design of a new Smart RADAR-Camera Perception System

	14h30 – 14h50	H. RIHAB: Normal Rank E-Order Interval Observer Design for Continuous Linear Time-Invariant Systems with Unknown Input and Bounded Disturbances
	14h50 – 15h10	A. GHANI: A Hierarchical Adaptive Observer Approach for Synchronization in Heterogeneous Multi-Agent Systems.
15h10-15h35	COFFEE BREAK	
15h35-16h35	IRSEEM's Platforms Visit (R. Khemmar, L. Lecrosnier, J. Fourre et C. Petrolesi)	
16h35–17h	Conclusion (R. Khemmar & A. Zemouche)	

Tuesday, 16 June 2026

9h-9h30	Final Event's Coffee Reception and Registration	
9h30-10h50	**CRAN Partner's Presentation**	
	9h30 - 9h50	Q.H. Nguyen: Resilient Trust-Aware Distributed Observer Design for Connected Vehicle Platoons
	9h50 - 10h10	S. Meng: Distributed Observer Design in the Application to Vehicle Platoons
	10h10 - 10h30	W. Jacques: Image-based Sensor Fusion and Estimation for use Localization, Tracking and Control of Automotive Systems
	10h30 - 10h50	Qi LI, Shengya MENG: Distributed High-Gain Observer Design Interconnected Nonlinear Systems for Vehicle Platoon Application
10h50-11h20	COFFEE BREAK	
11h20-12h15	**IBISC Partner's Presentation**	
	11h20 – 11h40	S. Ahmedali : Observateur adaptatif pour systèmes non linéaires à mesures échantillonnées : application aux véhicules.
	11h40 – 12h	S. Ahmedali: Data-Driven Observer Design for Nonlinear Vehicle Dynamics using Deep Radial Basis Function Networks with Delay Compensation.
12h - 14h	Lunch Break	
14h - 14h45	**FAAR Partner's Presentation**	
	A. ALIF : Détection, localisation et suivi d'une cible mobile à base de capteurs de perception, d'IA et d'un estimateur non linéaire	
14h45 – 15h15	COFFEE BREAK	
15h15 – 16h15	Round Table	
16h15 – 17h	Final EVENT Conclusion (R. KHEMMAR and A. ZEMOUCHE)	

ARTISMO Abstract.

The development of controllers with high performance and reliability for autonomous and connected vehicles will require real-time measurements or estimates of many variables on each vehicle. Examples of variables that are needed for feedback include longitudinal distances, velocities, and accelerations of other nearby vehicles; lateral position of the vehicle in its own lane; vehicle yaw angle; slip angle; yaw rate; steering angle; lateral acceleration; and roll angle. There are also environmental variables that need to be measured, such as tire-road friction coefficient, snow cover on the road, and the presence of unexpected obstacles. Measurement of all of the above variables requires significant expense. Indeed, some of the sensors above, such as slip angle and roll angle, can be extremely expensive to measure, requiring sensors that cost thousands of dollars. For example, the Datron optical sensor for measurement of slip angle has a price over 10ke. In addition, several variables cannot be measured due to the unavailability of sensors (at any cost). Examples include positions and accelerations of cars that are further upstream (e.g., the lead car of a platoon). Only the position of the immediately preceding car ahead can currently be measured. Furthermore, autonomous and connected vehicles require highly reliable sensors and actuators. Failure of any one sensor or actuator, due to faults, cyber-attacks or denial of service, can cause a disastrous accident. Hence, reliable fault diagnostic and fault handling systems are also needed. Such systems cannot be based on hardware redundancy, which requires many extra copies of the same sensors. Instead, they need to rely on estimation algorithms and analytical redundancy. For all the above reasons, the development of intelligent estimation algorithms is highly important for autonomous vehicles.

Partners

- FAAR SAS.
- CRAN – Centre de Recherche en Automatique de Nancy.
- IBISC - Informatique, Bioinformatique, Systèmes Complexes.
- IRSEEM – Institut de Recherche en Systèmes Electroniques Embarqués.

Dates.

Beginning and duration of the project: February 2021 – 48 Months – Extended until June 2026

Financial.

- Project ANR 20-CE48-0015
- Grant: 514.283€

CRAN's Presentations

Title: *Resilient Trust-Aware Distributed Observer Design for Connected Vehicle Platoons*

Authors: *Q.H. Nguyen, S. Meng, M. Haddad, H. Rafaralahy, and A. Zemouche*

Summary:

This paper introduces a trust-aware distributed observer designed to maintain accurate state estimation for connected vehicle platoons, even when the system is under cyberattack. The framework evaluates the reliability of shared data through a behavioral divergence metric, which then dynamically adjusts the observer's weighting gains to prioritize trustworthy information.

- **Trust Score Framework:** The system uses a multi-layer approach including Local Trust (physical consistency of sensors), Distributed Trust (consistency between global estimates), and Target Trust (a combination of both).
- **Observer Architecture:** It utilizes two layers: a Local Observer (LO) for the vehicle's own state and a Distributed Observer (DO) that fuses neighbor data.
- **Resilience:** The design is proven stable via Lyapunov analysis and is effective against bias, replay, and Denial-of-Service (DoS) attacks.
- **Simulation Results:** Tests on a four-vehicle platoon showed that the system successfully identifies communication disruptions and restores confidence once attacks cease.

Bios of presenters:

Quang Huy NGUYEN: PhD Candidate in Automatic Control at the University of Lorraine, France. He holds an M.Sc. and a Bachelor's degree in Control Science and Engineering from the University of Aix-Marseille, France. Experience: He serves as a temporary teacher at the IUT de Longwy, teaching subjects such as Automatics, Industrial Computing, and Embedded Systems. Research Interests: His work focuses on intelligent transportation systems, specifically cyberattack detection and estimation using adaptive sliding mode differentiators and discrete-time observers.

Title: *Distributed Observer Design in the Application to Vehicle Platoons*

Authors: *Shengya Meng, Ali Zemouche, Marouane Alma*

Summary: A vehicle platoon consists of multiple connected and autonomous vehicles that cooperate through inter-vehicle communication to improve traffic safety, efficiency, and energy consumption. In such systems, a distributed observer enables each vehicle to estimate the states of the entire platoon using only local measurements and information received from neighboring vehicles.

As part of our research on distributed estimation for connected autonomous vehicles, we have developed several distributed observer designs to address different practical challenges. First, a distributed unknown input observer is proposed to improve robustness against sensor faults, such as LiDAR failures, allowing reliable state estimation. Second, a distributed high-gain observer is designed to handle nonlinear longitudinal vehicle dynamics, where each vehicle reconstructs the global platoon states under nonlinear system behavior. These observer designs provide each vehicle with a global perception of the platoon, which is crucial for distributed control.

Based on this motivation, this work further develops a distributed observer-controller co-design framework to guarantee string stability under external disturbances. The observer estimates the tracking errors of all vehicles with respect to a global desired trajectory, and these estimates are directly used for control input generation. The method removes the need for explicit local spacing calculations and avoids requiring access to the private control inputs of other vehicles. By combining observer robustness and string stability into a unified weighted H_∞ performance index, the observer and controller are designed simultaneously. The approach is validated in QLabs with four QCar2 vehicles, demonstrating effective string-stable platooning under smooth-road, wavy-road, and start-stop scenarios.

Bios of presenters:

Shengya Meng is currently pursuing her Ph.D. degrees under the supervision of Ali Zemouche and Marouane Alma at the Centre de Recherche en Automatique de Nancy (CRAN UMR CNRS 7039) at Université de Lorraine, funded by the China Scholarship Council, since September 2023. She received her B.S. degree in automation from Guizhou University, Guiyang, China, in 2020, and her M.S. degree in control theory and control engineering from Northeastern University, Shenyang, China, in 2023. Her research interests included distributed control, distributed observer design, and their applications to real-world models.

Title: *Image-based Sensor Fusion and Estimation for use in Localisation, Tracking and Control of Automotive Systems*

Authors: *W. Jacques, H. Bessafa, Z. Belkhatir, A. Zemouche, R. Rajamani*

Summary: Advanced Driver Assistance Systems (ADAS) play a critical role in enhancing vehicle safety by supporting the driver in maintaining secure and safe control of their vehicle. This talk discusses the uses of estimation systems, sensor fusion, and computer vision for ADAS. This will be structured in two systems. The first system looks at the integration of visual odometry with a state-space high-gain observer for robust vehicle localisation. An Extended Kalman Filter enhances the vehicle's path localisation by fusing GNSS data with the visual odometry and the high-gain observer. The second system proposes a low-cost, vision-based Adaptive Cruise Control (ACC) framework that relies solely on a monocular camera for vehicle detection, tracking, and control. The method integrates a model-based high-gain observer with the YOLO computer-vision algorithm to estimate surrounding vehicles' trajectories directly from the ego-vehicle camera frames. These estimates are then incorporated into an MPC scheme to achieve real-time ACC functionality. We will show the performance of both systems using CARLA simulator's data.

Bios of presenters:

William Jacques received his M.Eng. degree in Electronic Engineering with Computer Systems from the University of Southampton in 2025. He is currently pursuing a Ph.D. at the School of Electronics and Computer Science, University of Southampton, United Kingdom. His research interests include state estimation, computer vision, and control for application in land- and air-based intelligent transportation systems.

Zehor Belkhatir received her Engineering and M.Sc. degrees in automatic control from Ecole Nationale Polytechnique, Algiers, Algeria, in 2012, and her Ph.D. degree in Electrical Engineering from King Abdullah University of Science and Technology, Jeddah, Saudi Arabia, in 2018. From 2018 to 2020, she held a Postdoctoral Research Scholar Position with the Department of Medical Physics, Memorial Sloan Kettering Cancer Center, New York. She is currently an Assistant Professor with the School of Electronics and Computer Science, University of Southampton, United Kingdom. Her research interests encompass work across the fields of applied mathematics, control systems theory, and data analysis. She is particularly interested in developing control, estimation and computational data analytics techniques with a particular emphasis in applying them to biomedical/biological systems and automotive/robotic systems. She is a member of the Technical Conferences Editorial Board (TCEB) and the European Control Association (EUCA) Conference Editorial Board.

Title: *Distributed High-Gain Observer Design of Interconnected Nonlinear Systems for Vehicle Platoon Application*

Authors: *Qi LI, Shengya MENG, Fanwei MENG, Cédric DELATTRE, Ali ZEMOUCHE*

Summary: State estimation is essential for the reliable control of large-scale interconnected multi-agent systems, particularly in vehicle platooning where cooperative maneuvers and collision avoidance are critical. Due to the triangular dependency inherent in platoon formations, vehicle dynamics are highly coupled, creating complex estimation error dynamics that are traditionally challenging to analyze. This work addresses this problem by proposing a systematic, distributed high-gain observer design. Our approach utilizes an error decomposition strategy, separating the dynamics into average and consensus components to effectively manage the coupling. The central innovation is a set of sufficient LMI (Linear Matrix Inequality) conditions that transform these complex coupled dynamics into a tractable convex optimization problem. This enables a constructive design procedure that yields all observer gains simultaneously while ensuring exponential convergence of the estimation error. The effectiveness of the proposed methodology is validated through simulations on a nonlinear vehicle platoon model, demonstrating its robust performance and relevance for practical automotive applications.

Bios of presenters:

Qi Li received his M.Eng. degree in Control Engineering from Northeastern University, China, in 2025. He is currently pursuing a Ph.D. at the CRAN (Centre de Recherche en Automatique de Nancy), University of Lorraine, France. His research interests focus on state estimation, distributed control theory, and their applications in intelligent transportation systems and autonomous vehicles.

IBISC’s Presentations

Titre : *Observateur adaptatif pour systèmes non linéaires à mesures échantillonnées : application aux véhicules*

Auteurs : *A. Benoudiba, N. Ait-oufroukh, S. Ahmed-ali, D. Ichalal*

Résumé : Les observateurs adaptatifs continus-discrets permettent l’estimation simultanée des états et des paramètres de systèmes non linéaires lorsque les sorties ne sont disponibles qu’aux instants d’échantillonnage. Cette présentation détaille une approche robuste fondée sur la combinaison d’un observateur d’état, d’un estimateur de paramètres et d’un prédicteur inter-échantillon, permettant de reconstruire la dynamique du système entre deux mesures successives. L’analyse de la dynamique couplée des erreurs, s'appuyant sur le théorème du petit gain, permet d’établir une propriété de stabilité de type ISS ainsi que des conditions moins conservatrices sur la période maximale d’échantillonnage admissible. La méthode est enfin illustrée sur un modèle de dynamique longitudinale de véhicule, avec des résultats montrant une estimation précise des états et des paramètres malgré les contraintes inhérentes aux mesures échantillonnées.

Biographies des auteurs :

Amayas BENOUDIBA a obtenu son diplôme de master en Smart Aerospace and Autonomous Systems à l’Université Paris-Saclay, en France, en 2023. Il prépare actuellement son doctorat au sein du laboratoire IBISC de l’Université d’Évry Paris-Saclay, en France. Ses travaux de recherche portent sur la conception d’observateurs adaptatifs pour l’estimation d’états et de paramètres, ainsi que sur les systèmes non linéaires sous contraintes d’échantillonnage, avec des applications dédiées à l’estimation et au contrôle de trajectoire pour les flottes de véhicules autonomes communicants

Naima Ait-Oufroukh a obtenu son diplôme de master en robotique en 1999, puis son doctorat en robotique et en traitement des données à l’Université d’Évry, en France, en 2002. Elle est actuellement maître de conférences à l’Université d’Évry. Ses travaux de recherche, menés au sein du laboratoire IBISC (Informatique, BioInformatique et Systèmes Complexes), portent sur l’assistance aux personnes en situation de handicap, notamment la localisation de robots, les capteurs, le traitement de données, l’aide à la conduite et les réseaux de neurones.

Sofiane Ahmed-Ali a obtenu sa licence en génie électrique de l’Université des sciences et de la technologie Houari-Boumédiène à Alger en 2001, ainsi que ses diplômes de master et de doctorat en génie électrique et en informatique de l’Université du Havre, en France, respectivement en 2004 et 2008. Après avoir rejoint l’ESIGELEC à Rouen en 2010 en tant que professeur associé, il est actuellement maître de conférences à l'Université d'Évry au sein du laboratoire IBISC. Ses intérêts de recherche portent sur la commande par modes glissants, les observateurs non linéaires et la commande tolérante aux fautes pour les dispositifs mécatroniques.

Dalil Ichalal a obtenu son diplôme de master à l'Université Paul Cézanne Aix-Marseille 3 (France) en 2006, puis son doctorat à l'Institut National Polytechnique de Lorraine (France) en 2009. En 2010, il a rejoint le département de génie électrique de l'Université d'Évry-Val-d'Essonne (France) et le laboratoire IBISC en tant que maître de conférences, et est actuellement professeur à l'Université d'Évry. Ses travaux de recherche portent sur l'estimation d'état et la conception d'observateurs, ainsi que sur le diagnostic et la commande tolérante aux fautes des systèmes non linéaires, avec des applications aux véhicules à deux et quatre roues ainsi qu'aux drones.

IRSEEM's Presentations

Title: *Perception-based AI for Smart Mobility.*

Authors: *R. KHEMMAR.*

Summary: Perception systems are essential for smart mobility applications, as they enable autonomous vehicles to understand complex and dynamic driving environments. This talk focuses on deep learning and multi-task learning approaches for improving scene understanding in autonomous driving. It presents a monocular multi-task perception framework

Bios of presenters: Radouane Khemmar graduated from the University of Poitiers in 2002 with an engineering degree in electronics, a post-graduate diploma (DEA) in industrial computing and a European Master's degree in image processing. He completed a PhD in image processing and computer vision at the University of Strasbourg in 2005. He then obtained a Habilitation à Diriger des Recherches (HDR) in computer vision, artificial intelligence and smart mobility from the Université Rouen Normandie in 2022. He worked.

Title: *Perception of Complex Scenes for Smart Mobility based on Deep Learning and Multi-Task Learning (MTL)*

Authors: *F. Jendoubi, R. Khemmar, R. Rossi, M. Haddad,*

Summary: Perception systems are essential for smart mobility applications, as they enable autonomous vehicles to understand complex and dynamic driving environments. This talk focuses on deep learning and multi-task learning approaches for improving scene understanding in autonomous driving. It presents a monocular multi-task perception framework that jointly performs 3D object detection and drivable area segmentation within a unified architecture to improve computational efficiency and environment understanding. It also introduces a multi-modal perception framework that combines camera and LiDAR data to enhance spatial representation and improve robustness in complex driving scenarios. In addition, a temporal perception approach is proposed to exploit sequential information from previous observations in order to better capture dynamic scene evolution and improve the understanding of moving objects over time. The performance of these approaches is evaluated on autonomous driving datasets to demonstrate their effectiveness for smart mobility applications.

Bios of presenters: **Jendoubi Firas** received his Engineering degree in Information and Communication Technologies (ICT), with a specialization in Artificial Intelligence, from Higher School of Communications of Tunis (SUP'COM) in 2023. He is currently pursuing a CIFRE Ph.D. in collaboration with IRSEEM, ESIGELEC, and SEGULA Technologies. His research focuses on complex scene analysis and understanding through deep learning, with a particular emphasis on perception systems for smart mobility. His research interests include computer vision, deep learning, object detection, multi-object tracking, and scene understanding for intelligent transportation systems.

Title: *Normal Rank $\mathcal{E}$-Order Interval Observer Design for Continuous Linear Time-Invariant Systems With Unknown Input and Bounded Disturbances*

Authors: R. E. H. Thabet, S. Ahmed Ali, V. Puig

Summary: In this talk, we present a new method to construct an interval observer for continuous-time LTI systems. Usually, the design of such observers is based on the cooperative property of the considered system dynamics, which is hard to satisfy in many cases. To overcome this issue, in a recent work, it has been shown that under some assumptions, the cooperativity of the studied system can be ensured by means of a time-

varying change of coordinates. However, a constructive method for the design of the observer gain, making the observation error dynamics simultaneously positive and stable, is still missing and remains an open problem, especially in the context of unknown input. In this paper, a constructive approach is provided to obtain not only the observer gain but also a new gain that cancels the unknown input of the system and reduces the effect of the bounded disturbances. The proposed approach also allows for estimating the bounds of the unknown input. The efficiency of the proposed methodology is shown through numerical simulations and is compared with simulations resulting from a new method proposed in a recent work.

Bios of presenters:

Rihab El Houda Thabet obtained the Engineering degree in 2010 and the M.Sc. degree in automatic and intelligent techniques in 2011 from National Engineering School of Gabes (Tunis). She received the Ph.D. degree from the University of Bordeaux in 2014. She was a Research Engineer at IRSEEM, Rouen from 2015 to 2016 and from 2018 to 2020. Since 2021, she rejoined ESIGELEC as a Teaching and Research Assistant Professor. Her research focuses on set-membership techniques for diagnosis and robust control of uncertain systems.

Sofiane Ahmed Ali received the B.Sc. degree in electrical engineering from the University of Technology Houari Boumedienne, Algiers, Algeria, in 2001, and the M.Sc. and Ph.D. degrees in electrical and computer engineering from the University of Le Havre, Le Havre, France in 2004 and 2008, respectively. In 2008, he was appointed as a Research and Development Engineer with the car manufacturer Renault. In 2010, he joined ESIGELEC, Rouen, France, as a Teaching and Research Assistant Professor and since September 2022, he is an Associate Professor with Paris-Saclay Val d'essone University, Paris, France, and the IBISC Laboratory, Paris, France. His current research interests include sliding mode control, nonlinear observers and fault tolerant control, and diagnosis in the field of mechatronics devices.

Vicenç Puig received the B.Sc./M.Sc. degree in telecommunications engineering and the Ph.D. degree in automatic control, vision, and robotics from the Universitat Politècnica de Catalunya (BarcelonaTech) (UPC), Barcelona, Spain, in 1993 and 1999, respectively. He is currently a Full Professor with the Automatic Control Department, UPC, and a Researcher with the Institut de Robòtica i Informàtica Industrial (IRI), CSIC-UPC. He is also the Director of the Automatic Control Department and the Head of the Research Group on Advanced Control Systems (SAC), UPC.

FAAR’s Presentations